

Buffers

How buffers work and the Henderson-Hasselbalch equation

Name _____ Class _____ Date _____

DATA

$$\text{pK}_a \text{ acetic acid} = 4.74 \quad \text{pK}_a \text{ NH}_4^+ = 9.25 \quad \text{pH} = \text{pK}_a + \log \frac{[\text{A}^-]}{[\text{HA}]}$$

Give pH to two decimal places.

A. Understanding

1. What is a buffer, and what two kinds of substance must it contain?

2. Which of these mixtures are buffers? Explain each answer briefly.

(a) $\text{HCl} + \text{NaCl}$ (b) $\text{CH}_3\text{COOH} + \text{CH}_3\text{COONa}$ (c) $\text{NH}_3 + \text{NH}_4\text{Cl}$ (d) $\text{NaOH} + \text{NaCl}$

3. Write equations to show how an acetic acid/acetate buffer removes (a) added H_3O^+ and (b) added OH^- .

B. Buffer pH

4. Calculate the pH of a buffer containing 0.10 M acetic acid and 0.15 M sodium acetate.

5. Calculate the pH of a buffer containing 0.20 M acetic acid and 0.10 M sodium acetate.

6. Calculate the pH of a buffer containing 0.20 M ammonia and 0.10 M ammonium chloride.

C. Adding acid

7. 0.500 L of buffer contains 0.20 M acetic acid and 0.20 M sodium acetate. Calculate the pH after adding 0.020 mol of HCl.

D. Challenge

8. What ratio $[\text{CH}_3\text{COO}^-]/[\text{CH}_3\text{COOH}]$ is needed to make an acetate buffer with pH 5.00?